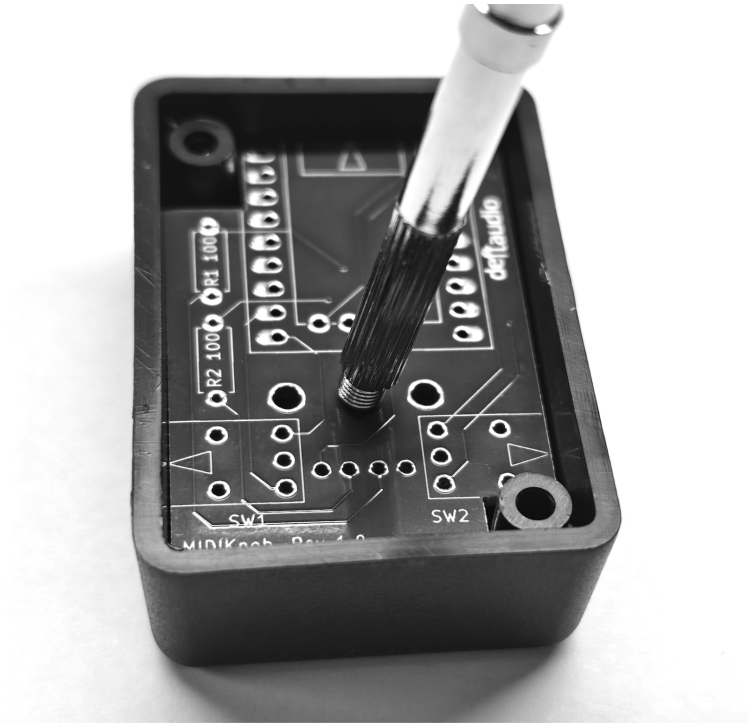
MIDIKnob Basic
Assembly Manual 1.0

Overview

The MIDIKnob Basic is a compact, yet powerful, programmable encoder controller designed for both MIDI control and everyday system interaction.
This is a **medium-difficulty kit** that requires soldering and firmware programming. Drilling holes in the enclosure are also required. The average assembly time is **25–30 minutes**.

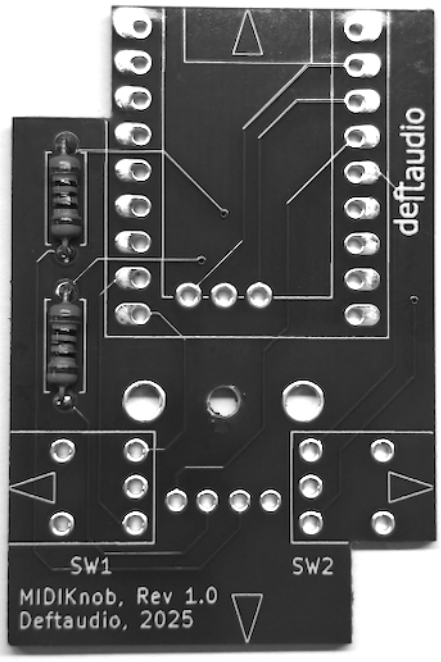
Hardware Assembly

Start by inspecting your package and verifying the received parts against the Bill of Materials (BOM) list.

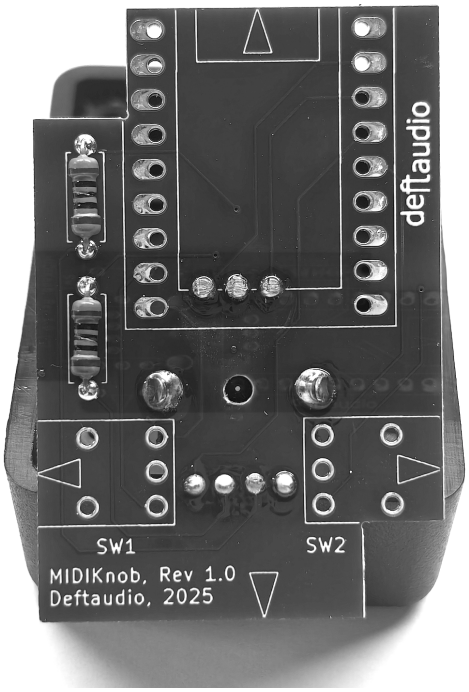


Follow these steps for hardware assembly:

Populate Resistors: Solder the two resistors (100 Ohm) onto the PCB.

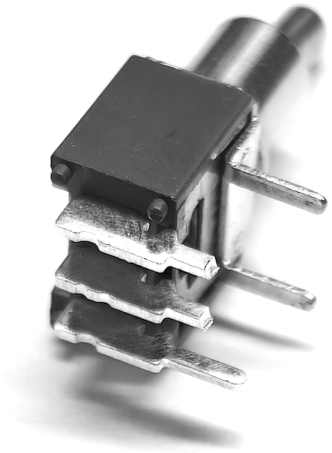


Solder Encoder: Solder the encoder from the bottom side of the board.

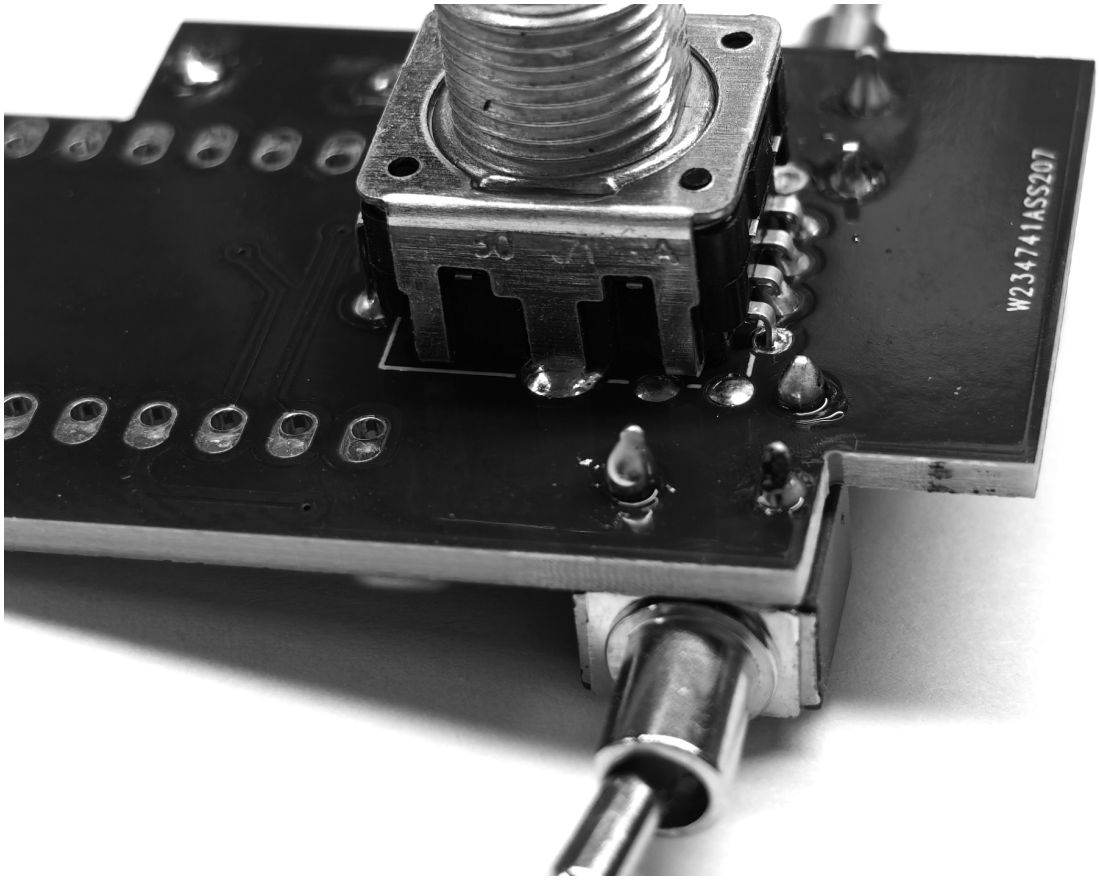


Two legs of switches need to be cut to prevent a circuit short.

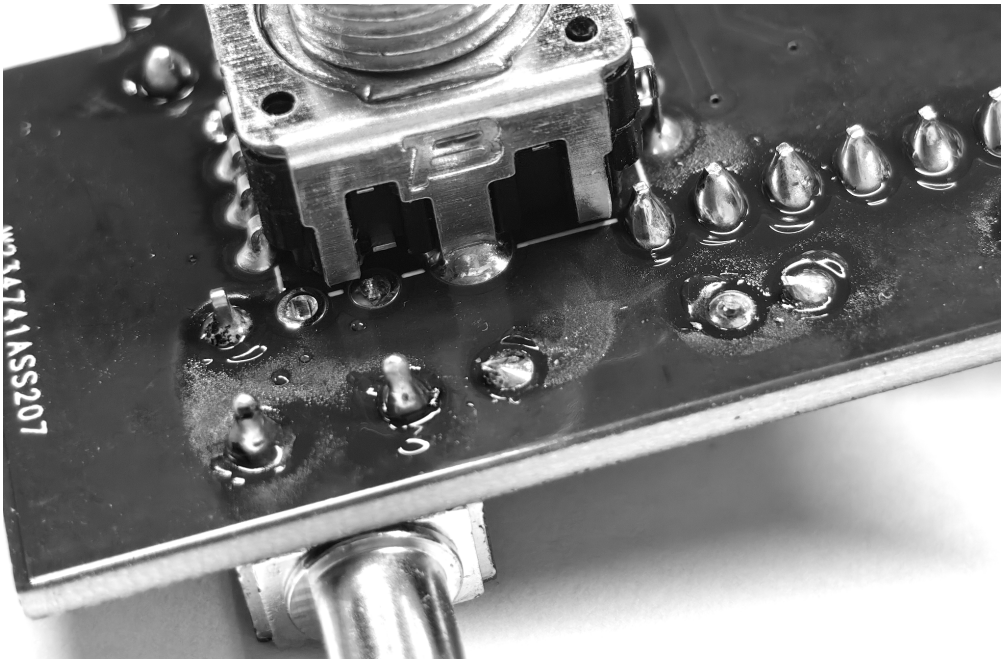
SW1 is ON-OFF-ON type and SW2 is ON-ON



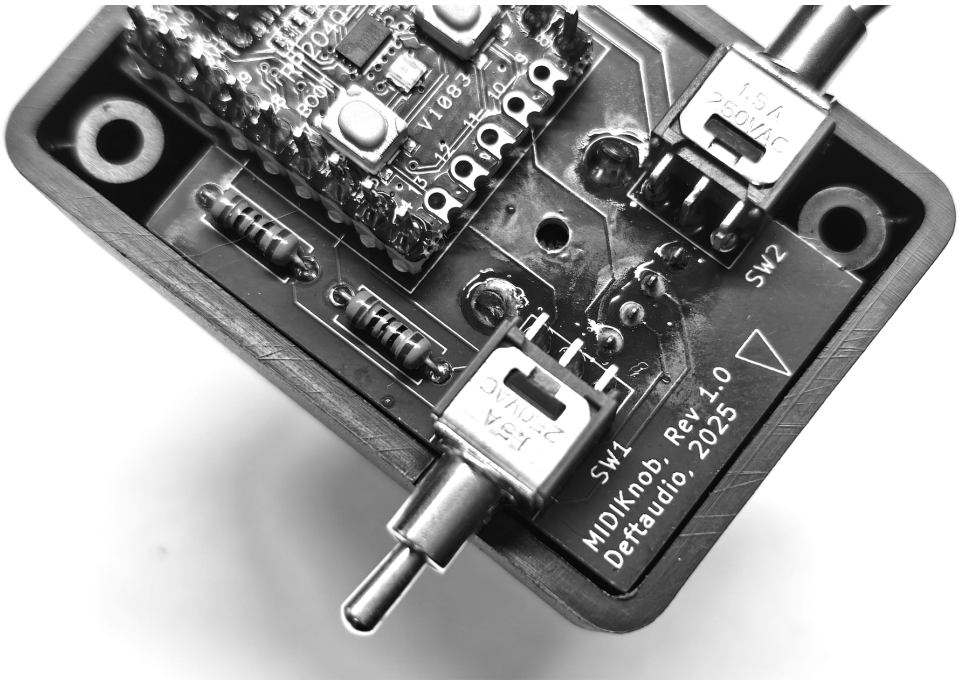
When installed on the board, switch legs shouldn’t touch the metal case of the encoder, see below:



Check Encoder Clearance: Ensure the encoder case does not short any pins on the bottom side.



Solder PR2040 Pico controller



Drill holes in the case:





Install Light Guide: Finally, insert the clear plastic light guide. Heat it up with a lighter or a heat gun and bend it to the shape shown below.


RP2040-Zero
deftaudio
SW1
SW2
MIDIKnob, Rev 1.0

Firmware Programming

The MIDIKnob family runs on **CircuitPython**, a real-time Python environment specifically designed for the Raspberry Pi Pico (RP2040). CircuitPython is ideal for rapid prototyping and is easy to learn. The MIDIKnob code is open-source Python under the BSD-3 license, and contributions are highly encouraged.
To deploy the firmware, you must first install CircuitPython on a blank RP2040 and then copy over the Python code files.

Installation Steps:

- **Download CircuitPython:** Download the latest CircuitPython UF2 file for the RP2040_zero board from https://circuitpython.org/board/waveshare_rp2040_zero/.
- **Connect Device:** Connect the MIDIKnob to your PC or Mac. It will appear as a USB flash drive.
- **Install UF2:** Copy the downloaded UF2 file onto the USB flash drive. The device will automatically self-reset and reappear as a new USB flash drive containing system files.
- **Download Firmware:** Download the MIDIKnob Firmware and unpack the zip archive from https://github.com/Deftaudio/Midi-boards/tree/master/MidiKnob/FW.
- **Copy Firmware Files:** Replace the content of the Pico USB drive with the content of the unpacked archive.
- **Finalize:** Safely disconnect and reconnect the MIDIKnob.

Once reconnected, the MIDIKnob code will be active and running. You will see the new startup sequence, which triggers various LED colors. However, the device will lack configuration in its non-volatile memory, so basic MIDI programming is needed.

Operating Modes:
The device supports three operating modes, selectable via the switch on the right side:

- **MIDI Mode**
- **System Volume**

In each mode, the switch on the left allows you to select one of **three stored presets**, providing instant access to multiple configurations.

MIDI Mode

In MIDI Mode, rotating the encoder sends a **MIDI CC (Control Change) message** on the assigned controller number. Pressing the encoder sends a **Note On / Note Off** message.
The device also responds to incoming **MIDI Clock**, causing the encoder's LED to pulse on the start of each bar and on every beat, providing clear visual timing feedback.

MIDI Programming: Programming MIDI behavior is quick and intuitive: simply **double-tap the encoder** and send a MIDI CC or Note On message from an external controller. The assignment is automatically saved to **non-volatile memory** and recalled upon power-up.

System Volume / Keyboard Mode

In System Volume mode, the device operates as a **USB HID (Human Interface Device) controller**, providing native control of your computer's audio volume. Rotating the encoder adjusts the volume up or down. Pressing the encoder sends a default keyboard command, which can be remapped as needed.

Customizing Firmware

To make changes to the device's behavior, you can edit the Python code:

1. **Open `Code.py`:** Use an external editor to open the `Code.py` file on the MIDIKnob USB drive.
    - **Recommendation:** Thonny is a simplistic editor that allows for easy changing, restarting, and configuring of the Pico board.
2. **Make Basic Changes:** Simple modifications, such as changing the keyboard shortcuts the MIDIKnob sends, involve finding the relevant line in the code and modifying it.

If you are not comfortable programming yourself, you can use an AI coding assistant such as Codex from ChatGPT. It's free for small projects. Start a new project, point the assistant to the MIDIKnob files, and ask it to make the changes for you. This approach works perfectly for straightforward code modifications and gives an immediate result.